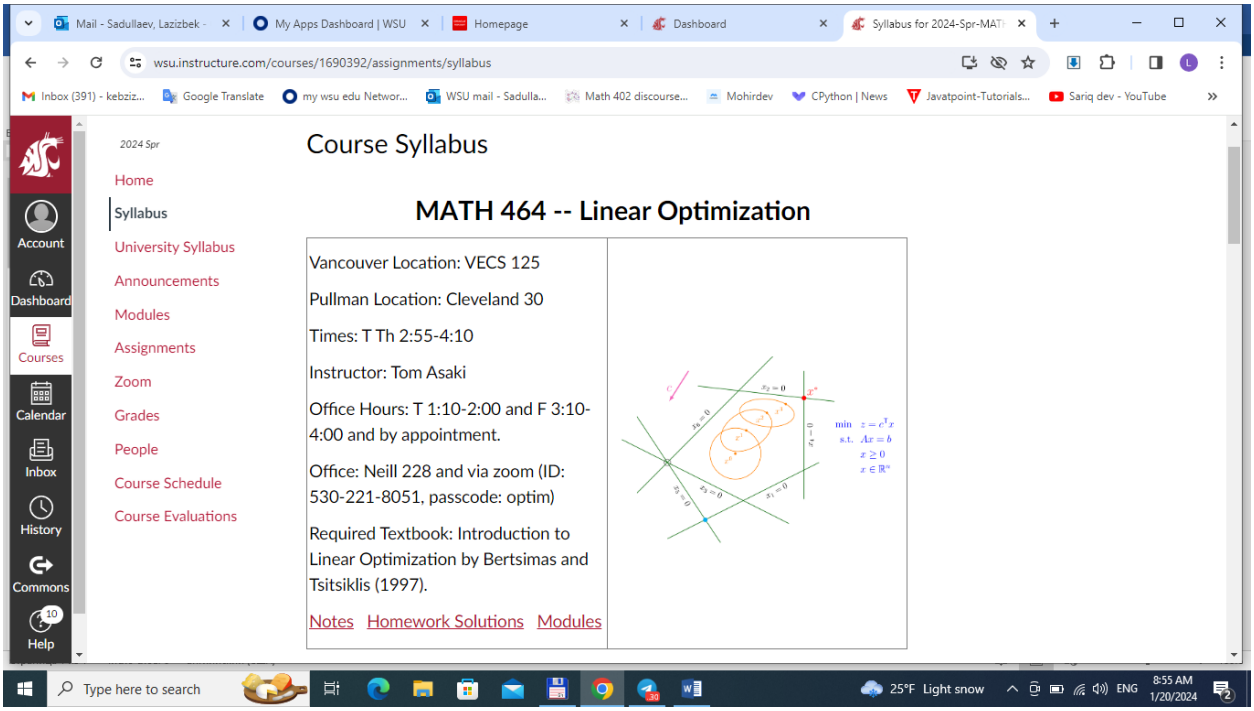

Course Syllabus

MATH 464 -- Linear Optimization

Vancouver Location: VECS 125 Pullman Location: Cleveland 30 Times: T Th 2:55-4:10 Instructor: Tom Asaki Office Hours: T 1:10-2:00 and F 3:10-4:00 and by appointment. Office: Neill 228 and via zoom (ID: 530-221-8051, passcode: optim) Required Textbook: Introduction to Linear Optimization by Bertsimas and Tsitsiklis (1997). Notes Homework Solutions Modules	

Description. This course explores the theory and application of linear programming -- a very important and practical sub-field of mathematical optimization. We will learn how mathematicians express optimization problems, model real-world decision-making problems, understand and practice solution methods, employ software for linear programs. This course is a mathematics Capstone course which is accessible to students from a variety of majors, provided that their mathematical background is solid. The course prerequisite is Math 273 or Math 283. However, completion of Math 301, Math 364 and/or Math 220/230/420 will be very helpful.

Graded 464 Skills.

- Model realistic decision making questions as mixed-integer programs. [Midterm 1, Project, Homework]
- Rigorously prove geometric claims related to linear programming. [Midterm 2, Homework]
- Construct clear justifications of algorithmic and geometric properties of linear programs. [Midterm 2, Homework]
- Solve small linear programs by hand using the Simplex Method and sensitivity analysis. [Midterm 2, Homework]
- Employ software to solve both small and large mixed-integer programs. [Homework, Project]
- Model and solve a larger real-world mixed-integer program. [Project]
- Construct a detailed final project report. [Project]

Grade Basis. Assessment of student achievement will be measured through homework assignments (20%), two examinations (40%), and the capstone project (40%). Class attendance is expected and could be used to justify small grade increases in borderline cases. Grades will be assigned using the following percentage scale: (A:90) (B+:85) (B:80) (C+:75) (C:70) (D:60).

Homework. Due to the large class size, and in an effort to provide meaningful and timely feedback, most assignments will not be collected, though solutions will be provided. Some assignments will be collected and graded. It is very important that you impose upon yourself a schedule of completing all assignments so that you are prepared for exams, classwork, and the project. The two lowest homework scores will be dropped in the final grade calculation.

In addition, submitted homework should be addressed with care, as full credit requires attention to detail. Each assignment is to be submitted on Canvas as a PDF document (LaTeX typeset documents are worth a small amount of extra credit). Submitted files should not be larger than about 1MB (for example, documents containing full resolution

cell-phone images are not acceptable). Consider these examples Download examplesof well-written homework solutions - please read them and apply the principles to your own work.

Exams. Two in-class midterm exams will be given - currently scheduled for February 15 and April 9. Each exam will be closed book/notes. Each will cover a specific subset of the course material. Each exam will be designed to test mathematical fluency, application skills, and topical understanding.

Project. Individual projects will be started and completed during the final three weeks of the semester and serve as the final examination for the course. The details of the project will be introduced on April 2 in class and will be due on April 29 (at the scheduled time of the final exam). In the project the student will demonstrate mastery of the course material through application to a real-world (or realistic) optimization task. The project will demonstrate problem formulation, problem modeling, solutions through software implementation, interpretation and analysis of the results, and mathematical and practical context understanding. Additional details will be available at the time projects are finalized and posted.

Attendance. Class attendance and participation is expected. Be prepared to answer questions and contribute to discussions. Lecture recordings will not be generally available. However, I will post most instructor notes.

Late Work Policy. All assignments are due on or before the posted due date/time and any work submitted late is subject to a zero score. Late work may be accepted for credit in some circumstances for which the delay is generally beyond the control of the student or in cases of significant need. Timely communication with the instructor is expected in such cases. Busyness, excessive obligations, and scheduling conflicts are not acceptable reasons for late work. Plan ahead, work ahead.

Schedule. (See Course Modules.) My plan is to be present in Vancouver for class on at least three occasions. We will cover material corresponding the following sections of the textbook: 1.1-1.5, 2.1-2.4, 3.1-3.5, 4.1-4.3, 5.1, 9.1-9.5. After the second midterm exam we will be focused on the final project, and will not cover new textbook material.

Academic Integrity is the cornerstone of higher education. As such, all members of the university community share responsibility for maintaining and promoting the principles of integrity in all activities, including academic integrity and honest scholarship. Academic integrity will be strongly enforced in this course. Students who violate WSU's Academic Integrity Policy (identified in Washington Administrative Code (WAC) 504-26-010(3) and -404) will receive a failing grade on any relevant assignment (and possibly for the

course), will not have the option to withdraw from the course pending an appeal, and will be reported to the Office of Student Conduct.

Cheating includes, but is not limited to, plagiarism and unauthorized collaboration as defined in the Standards of Conduct for Students, WAC 504-26-010(3). You need to read and understand all of the definitions of cheatingLinks to an external site.. If you have any questions about what is and is not allowed in this course, you should ask course instructor before proceeding.
If you wish to appeal a faculty member's decision relating to academic integrity, please use the form available at conduct.wsu.eduLinks to an external site..